

Notes: Giroud & Mueller (JF, 2015)

Capital and Labor Reallocation within Firms

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
2.1	Contribution relative to internal capital markets literature	3
2.2	Contribution relative to financial constraints literature	3
2.3	Contribution relative to geography and monitoring literature	4
2.4	Contribution relative to productivity and resource reallocation literature	4
3	Data and measurement	4
3.1	Plant-level Census data	4
3.2	Treatment and other-plant indicators	5
3.3	Travel-time measurement	5
3.4	Main outcome variables	5
3.5	Productivity estimation	5
3.6	Financing constraints	6
4	Empirical specifications	6
4.1	Baseline plant-level difference-in-differences	6
4.2	Financing-constraint interaction	6
4.3	Dynamic event-study specification	6
4.4	Heterogeneity across other plants	7
4.5	Firm-level regressions	7
5	Identification and empirical design	7
5.1	Source of identifying variation	7
5.2	Role of fixed effects	7
5.3	Why the other-plant effect is especially informative	8
5.4	Main validity checks	8
6	Tables and figures	9
6.1	Table I: summary statistics	9
6.2	Tables II and III: plant-level resource allocation and productivity	9
6.3	Table IV: dynamics of treatment effects	10
6.4	Tables V–VII: financing constraints and robustness	11
6.5	Tables VIII and IX: which other plants are affected?	12
6.6	Tables X and XI: firm-level investment, employment, and productivity	13
6.7	Table XII: other sources of funding	14
7	Short summary	14
8	Conclusion	15
8.1	Core conclusion	15
8.2	Economic intuition	15

8.3	Takeaway	15
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1 Big picture

- **Question.** Do firms actively reallocate capital and labor across plants when one plant receives a positive shock to investment opportunities?
- **Core hypothesis.** Headquarters can create value through an internal capital market by moving scarce resources from lower-value uses to higher-value uses inside the firm.
- **Empirical shock.** The treatment is the introduction of a new airline route that reduces travel time between headquarters and a plant. This makes it easier for headquarters to monitor, advise, and transfer knowledge to the treated plant, increasing the appeal of investing there.
- **Main mechanism.** If the firm is financially constrained, expanding the treated plant requires resources. Headquarters therefore withdraws capital and labor from other plants in the same firm.
- **Key asymmetry.** Reallocation appears only for financially constrained firms. Financially unconstrained firms can fund expansion at the treated plant without reducing resources elsewhere.
- **Treated-plant result.** Investment, employment, TFP, and return on capital increase at treated plants after travel time to headquarters falls.
- **Other-plant result.** Other plants in financially constrained firms experience lower investment and employment, while other plants in unconstrained firms do not.
- **Efficiency result.** Headquarters primarily withdraws resources from other plants that are less productive, outside the firm's core industries, and geographically farther from headquarters.
- **Firm-level result.** At the firm level, investment and employment rise, and aggregate productivity rises. This supports the view that reallocation is value enhancing rather than pure lobbying or tunneling.
- **Takeaway.** Internal capital markets appear to operate efficiently when external finance is scarce: headquarters reallocates both capital and labor toward better investment opportunities.

2 Incremental contribution of the paper

2.1 Contribution relative to internal capital markets literature

- Earlier empirical work often uses Compustat segment data and industry Tobin's Q to infer internal capital allocation. Segment data are coarse and cannot observe establishments, employment, or plant-level productivity.
- This paper uses confidential U.S. Census plant-level data, allowing direct observation of capital, employment, shipments, productivity, and plant locations.
- Instead of asking whether a conglomerate segment responds to other segments' industry Q , the paper observes a plausibly exogenous shock to one plant and asks whether resources move out of other plants.
- The result is much closer to the theoretical thought experiment in Stein (1997): when project B becomes more attractive, does investment in project A fall?

Interpretation: The incremental contribution is not just better data. It is a cleaner test of the central reallocation mechanism behind the efficient internal capital markets hypothesis.

2.2 Contribution relative to financial constraints literature

- A large literature asks whether financing constraints affect investment, often by interacting cash flow or investment opportunities with firm-level constraint proxies.

- This paper shows a different implication of financial constraints: constrained firms must finance new opportunities internally, so their investment response creates negative spillovers on other internal projects.
- The paper uses multiple constraint measures: KZ index, WW index, SA index, credit ratings, debt-to-cash-flow, and investment in excess of cash flow.
- The key result survives all measures: negative spillovers are concentrated among financially constrained firms.

Interpretation: The paper turns financing constraints into a resource-allocation prediction. The question is not only whether a constrained firm invests less, but whether it reallocates scarce resources internally.

2.3 Contribution relative to geography and monitoring literature

- Giroud (2013) shows that shorter travel time between headquarters and plants improves plant performance.
- This paper builds on that shock but asks a different question: what happens to the rest of the firm?
- The answer is that the treated plant’s gain has within-firm spillovers only when the firm is financially constrained.
- The paper therefore connects geography, monitoring, financial constraints, and internal resource allocation.

Interpretation: The incremental contribution is to move from a local monitoring effect to a firm-wide reallocation effect.

2.4 Contribution relative to productivity and resource reallocation literature

- The paper shows that resource reallocation happens not only across firms or across industries, but also inside multi-plant firms.
- The reallocation is selective: headquarters reduces investment and employment more at plants whose opportunity cost is lower.
- Firm-level productivity increases, implying that within-firm reallocation can improve aggregate productive efficiency.

Interpretation: The paper gives micro evidence that headquarters can act as an internal planner, reallocating resources in a way that raises total firm productivity.

3 Data and measurement

3.1 Plant-level Census data

- The paper uses plant-level data from the U.S. Census Bureau’s Longitudinal Business Database and manufacturing data sources.
- The sample covers manufacturing plants from 1977 to 2005.
- The main sample is restricted to “pure” manufacturing firms so that all relevant operations are observed in manufacturing Census data.
- The data contain employment, shipments, capital expenditures, capital stock, materials, labor costs, plant age, plant location, and firm identifiers.
- The final plant-level sample contains 291,358 plant-year observations.

Interpretation: The plant-level panel lets the authors observe resource flows within firms directly, rather than inferring them from segment-level accounting data.

3.2 Treatment and other-plant indicators

- A plant is treated when a new airline route reduces travel time between the plant and headquarters.
- **Treated** equals one for the treated plant after the new route is introduced.
- **Other** equals one for other plants in the same firm after one of the firm’s plants has been treated.
- The control group includes plants not yet treated and plants not yet classified as other plants.

Interpretation: The distinction between treated and other plants is the core design. The treated plant receives the positive investment opportunity shock; other plants reveal whether headquarters reallocates resources internally.

3.3 Travel-time measurement

- Travel time is computed between headquarters ZIP code and plant ZIP code.
- The authors compare driving time with the fastest air itinerary.
- Air travel time includes driving to the origin airport, flight time, layover time, airport time, and driving from the destination airport to the plant.
- A new route is a treatment only if it reduces the previously fastest travel time.
- Route types include indirect-to-indirect, indirect-to-direct, direct-to-direct, and road-to-flight changes.

Interpretation: Travel time is preferable to distance because monitoring costs depend on the actual time required to visit a plant, not simply miles between locations.

3.4 Main outcome variables

- **Investment:** capital expenditures divided by capital stock, both measured in real 1997 dollars.
- **Employment:** log number of employees.
- **TFP:** residual from an industry-year production function.
- **ROC:** shipments minus labor and material costs divided by capital stock.
- Outcomes are industry-adjusted by subtracting the three-digit SIC industry-year median.
- Outcomes are winsorized at the 2.5th and 97.5th percentiles.

Interpretation: The outcomes separately capture resource quantities and resource efficiency. Investment and employment show reallocation; TFP and ROC show whether reallocation raises productivity.

3.5 Productivity estimation

TFP is the residual from a log-linear Cobb–Douglas production function estimated separately by industry and year:

$$y_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \beta_m m_{it} + \varepsilon_{it}, \quad (1)$$

where y is log output and k , l , and m are log capital, labor, and materials.

Interpretation: Estimating this equation separately by industry-year allows factor intensities to vary across industries and over time. TFP is therefore a within-industry productivity measure.

3.6 Financing constraints

- The main constraint measures are the Kaplan–Zingales (KZ) index and the Whited–Wu (WW) index.
- Firms are classified as financially constrained if their constraint measure is above the yearly median in the pre-treatment year.
- Firms below the median are classified as financially unconstrained.
- Robustness checks use the SA index, credit-rating status, debt-to-cash-flow ratio, and investment in excess of cash flow.

Interpretation: Using pre-treatment values avoids classifying firms based on outcomes that may themselves be affected by treatment.

4 Empirical specifications

4.1 Baseline plant-level difference-in-differences

The baseline plant-level regression is

$$y_{ijlt} = \alpha_i + \alpha_{lt} + \beta_1 Treated_{ijt} + \beta_2 Other_{ijt} + \gamma' X_{ijt} + \varepsilon_{ijlt}, \quad (2)$$

where i indexes plants, j firms, l plant locations, and t years.

- α_i are plant fixed effects.
- α_{lt} are MSA-by-year fixed effects.
- $Treated_{ijt}$ identifies treated plants after a route reduces travel time.
- $Other_{ijt}$ identifies other plants in the same firm after one plant is treated.
- X_{ijt} includes lagged plant size and lagged plant age.

Interpretation: β_1 measures the effect on the plant whose monitoring improves. β_2 measures the spillover on other plants in the same firm.

4.2 Financing-constraint interaction

To test whether reallocation is driven by financial constraints, the paper estimates

$$y_{ijlt} = \alpha_i + \alpha_{lt} + \beta_1 Treated_{ijt} \times FC_j + \beta_2 Treated_{ijt} \times NonFC_j \quad (3)$$

$$+ \beta_3 Other_{ijt} \times FC_j + \beta_4 Other_{ijt} \times NonFC_j + \gamma' X_{ijt} + \varepsilon_{ijlt}. \quad (4)$$

Interpretation: The key prediction is $\beta_3 < 0$ and $\beta_4 \approx 0$. Other plants should lose resources only when the firm is financially constrained.

4.3 Dynamic event-study specification

The dynamics regression replaces the single $Other \times FC$ effect with event-time interactions:

$$y_{ijlt} = \alpha_i + \alpha_{lt} + \sum_k \delta_k (Other_{ijt} \times FC_j \times Year(k)) + \dots + \varepsilon_{ijlt}. \quad (5)$$

Interpretation: The event-study checks whether spillovers occur after treatment and whether there are pre-treatment trends. The effect is strongest in the years immediately after treatment.

4.4 Heterogeneity across other plants

The cross-sectional regressions interact $Other \times FC$ with plant characteristics:

$$y_{ijlt} = \alpha_i + \alpha_{lt} + \beta_L Other_{ijt} \times FC_j \times Low_i + \beta_H Other_{ijt} \times FC_j \times High_i + \dots + \varepsilon_{ijlt}. \quad (6)$$

The characteristics are:

- low versus high plant productivity;
- peripheral versus main industry;
- far versus near to headquarters;
- firms with few versus many other plants.

Interpretation: Efficient reallocation predicts that headquarters withdraws more from plants with lower opportunity costs.

4.5 Firm-level regressions

The firm-level analogue is

$$y_{jt} = \alpha_j + \alpha_t + \beta Treatment_{jt} + \gamma' X_{jt} + \varepsilon_{jt}, \quad (7)$$

where $Treatment_{jt}$ equals one if any plant of firm j has been treated by year t .

The financing-constraint version is

$$y_{jt} = \alpha_j + \alpha_t + \beta_1 Treatment_{jt} \times FC_j + \beta_2 Treatment_{jt} \times NonFC_j + \gamma' X_{jt} + \varepsilon_{jt}. \quad (8)$$

Interpretation: Firm-level regressions ask whether plant-level reallocation is merely redistributive or raises aggregate firm-wide investment, employment, and productivity.

5 Identification and empirical design

5.1 Source of identifying variation

- Identification comes from staggered introductions of airline routes that reduce headquarters-to-plant travel time.
- Route introductions are plausibly exogenous to any one plant's investment because airline routes are determined by broader airline network decisions.
- Treatments occur at the headquarters-plant pair level, not merely at the plant location level.

Interpretation: The identifying variation is a reduction in monitoring cost for one headquarters-plant pair, not a general local shock to the plant's region.

5.2 Role of fixed effects

- Plant fixed effects absorb permanent differences across plants.
- MSA-year fixed effects absorb local economic shocks at the plant location.

- Because not all plants in an MSA have headquarters in the same location, treatment is identified within location-year cells.
- Standard errors are clustered at the firm level.

Interpretation: MSA-year fixed effects are especially important because airline routes may be introduced to places experiencing local booms.

5.3 Why the other-plant effect is especially informative

- Endogeneity concerns are strongest for treated plants because a local shock could affect both the treated plant and airline routes.
- Those concerns are weaker for other plants in the same firm located elsewhere.
- A negative effect on other plants of constrained firms is hard to explain with local demand shocks alone.

Interpretation: The spillover to other plants is the most direct evidence of internal resource reallocation.

5.4 Main validity checks

- The authors compare constrained and unconstrained firms.
- They use several alternative financing-constraint measures.
- They compare public and private firms.
- They test whether resources are taken more from low-productivity, non-core, and distant plants.
- They examine firm-level productivity to distinguish efficient reallocation from pure redistribution or lobbying.

Interpretation: The identification is strengthened because many independent predictions of the same internal-reallocation mechanism hold simultaneously.

6 Tables and figures

6.1 Table I: summary statistics

Read:

- The sample contains 291,358 plant-year observations from 1977 to 2005.
- Average plants have 410 employees, shipments of 97,255 thousand dollars, capital stock of 42,078 thousand dollars, and investment of 3,848 thousand dollars.
- Eventually treated plants and other plants are broadly similar to the full sample.
- The investment-to-capital-stock ratio is around 0.10 for all plants and treated plants.

Economic message: The treated, other, and remaining plants are not obviously different in raw means, supporting the comparability of the empirical design.

Table I
Summary Statistics

“Eventually Treated Plants” refers to plants that are treated during the sample period, that is, plants whose travel time to headquarters is reduced through the introduction of a new airline route. “Eventually ‘Other’ Plants” refers to plants that become “other” plants during the sample period, that is, plants that belong to the same firm as the treated plant. Shipments, capital stock, and investment are expressed in 1997 dollars (in 1,000s) using four-digit SIC deflators from the NBER-CES Manufacturing Industry Database. Capital stock is constructed using the perpetual inventory method. All figures are sample means. Standard deviations are in parentheses. The sample period is from 1977 to 2005.

	All Plants	Eventually Treated Plants	Eventually “Other” Plants	Remaining Plants
Employees	410 (929)	379 (756)	432 (968)	376 (975)
Shipments	97,255 (360,818)	95,403 (304,582)	103,929 (360,623)	79,235 (411,983)
Capital Stock	42,078 (141,084)	41,666 (139,036)	45,756 (147,219)	31,501 (122,738)
Investment	3,848 (53,589)	3,646 (15,735)	3,969 (21,316)	3,701 (113,504)
Investment / Capital Stock	0.10 (0.14)	0.10 (0.13)	0.10 (0.14)	0.11 (0.16)
Observations	291,358	61,007	172,667	57,684

6.2 Tables II and III: plant-level resource allocation and productivity

Read:

- Table II shows that treated plants increase investment by about 0.010 and employment by about 0.025 in the baseline specification.
- For financially constrained firms, other plants experience lower investment and employment: $Other \times FC$ is negative and significant.
- For financially unconstrained firms, other-plant effects are close to zero.
- Table III shows that treated plants experience higher TFP and ROC.
- Productivity at other plants does not fall much, suggesting headquarters avoids large productivity losses at the plants from which resources are withdrawn.

Economic message: The core mechanism appears in Table II: constrained firms fund treated-plant expansion by reducing resources at other plants. Table III shows that the treated plant becomes more productive.

Table II
Plant-Level Investment and Employment

The dependent variable is either plant-level investment (columns (1) to (3)) or plant-level employment (columns (4) to (6)). Investment is the ratio of capital expenditures to capital stock at the plant level. Employment is the natural logarithm of the number of employees of the plant. Both variables are industry-adjusted by subtracting the industry median across all plants in a given three-digit SIC industry and year. *Treated* is a dummy variable that equals one if a new airline route has been introduced that reduces the travel time between the plant and its headquarters. *Other* is a dummy variable that equals one if the plant belongs to the same firm as the treated plant and the treated dummy is set to one. *FC (Non-FC)* is a dummy variable indicating whether the plant belongs to a firm whose measure of financing constraints lies above (below) the median across all firms in the year prior to the treatment. In columns (2) and (5), financing constraints are measured using the KZ index of Kaplan and Zingales (1997). In columns (3) and (6), financing constraints are measured using the WW index of Whited and Wu (2006). The control variables are plant size and plant age. Size is the natural logarithm of the plant's shipments. Age is the natural logarithm of one plus the number of years since the plant has been in the LBD. Both variables are lagged. Standard errors are clustered at the firm level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Investment			Employment		
	(1)	KZ Index (2)	WW Index (3)	(4)	KZ Index (5)	WW Index (6)
Treated	0.010*** (0.001)			0.025*** (0.004)		
Other	-0.001 (0.001)			-0.002 (0.003)		
Treated × FC		0.008*** (0.002)	0.008*** (0.003)		0.019*** (0.006)	0.019** (0.008)
Treated × Non-FC		0.012*** (0.002)	0.011*** (0.001)		0.028*** (0.005)	0.026*** (0.004)
Other × FC		-0.002* (0.001)	-0.003 (0.002)		-0.006* (0.004)	-0.007 (0.006)
Other × Non-FC		0.000 (0.001)	-0.000 (0.001)		0.001 (0.004)	-0.000 (0.003)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Plant Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
MSA × Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	291,358	291,358	291,358	291,358	291,358	291,358
R ²	0.32	0.32	0.32	0.92	0.92	0.92

Table II: plant-level investment and employment

Table III
Plant-Level Productivity

This table presents variants of the regressions in Table II where the dependent variable is either plant-level TFP (columns (1) to (3)) or plant-level ROC (columns (4) to (6)). TFP is the estimated residual from a regression of the logarithm of output on the logarithms of capital, labor, and material inputs (see Section II.A). ROC is the value of shipments minus labor and material costs divided by capital stock and is industry-adjusted by subtracting the industry median across all plants in a given three-digit SIC industry and year. TFP is industry-adjusted by construction. All other variables are described in Table II. Standard errors are clustered at the firm level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	TFP			ROC		
	(1)	KZ Index (2)	WW Index (3)	(4)	KZ Index (5)	WW Index (6)
Treated	0.012*** (0.001)			0.014*** (0.002)		
Other	-0.000 (0.001)			-0.000 (0.001)		
Treated × FC		0.009*** (0.002)	0.007** (0.003)		0.010*** (0.002)	0.007** (0.003)
Treated × Non-FC		0.014*** (0.002)	0.013*** (0.001)		0.016*** (0.002)	0.016*** (0.002)
Other × FC		-0.001 (0.002)	-0.002 (0.002)		-0.001 (0.002)	-0.002 (0.003)
Other × Non-FC		0.000 (0.001)	0.000 (0.001)		0.000 (0.002)	0.001 (0.002)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Plant Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
MSA × Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	291,358	291,358	291,358	291,358	291,358	291,358
R ²	0.61	0.61	0.61	0.62	0.62	0.62

Table III: plant-level productivity

6.3 Table IV: dynamics of treatment effects

Read:

- The dynamic specification shows no strong pre-treatment spillover for other plants.
- Treated-plant effects for constrained firms appear after the route introduction and persist for several years.
- Other-plant effects for constrained firms become negative after the treatment, especially in years 1 and 2.
- The dynamic pattern is consistent with reallocation taking place after headquarters receives easier access to the treated plant.

Economic message: The event timing supports the reallocation interpretation: resources are not withdrawn from other plants before the investment opportunity improves.

Table IV
Dynamics of the Treatment Effect

This table presents variants of the regressions in Table II where *Other* $\times$ *FC* and *Other* $\times$ *Non-FC* are interacted with dummy variables indicating whether a plant-year observation is measured one year before the treatment (−1), in the year of the treatment (0), one, two, and three years after the treatment (1, 2, and 3, respectively), or four and more years after the treatment (4+). All other variables are described in Table II. Standard errors are clustered at the firm level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Investment		Employment	
	KZ Index (1)	WW Index (2)	KZ Index (3)	WW Index (4)
Treated $\times$ FC $\times$ Year(−1)	−0.003 (0.004)	0.002 (0.005)	−0.003 (0.011)	−0.002 (0.017)
Treated $\times$ FC $\times$ Year(0)	0.005 (0.003)	0.004 (0.006)	0.004 (0.010)	0.003 (0.014)
Treated $\times$ FC $\times$ Year(1)	0.014*** (0.003)	0.014*** (0.004)	0.031*** (0.010)	0.041*** (0.014)
Treated $\times$ FC $\times$ Year(2)	0.009** (0.003)	0.010* (0.005)	0.029*** (0.008)	0.030*** (0.012)
Treated $\times$ FC $\times$ Year(3)	0.008* (0.004)	0.008* (0.005)	0.019** (0.010)	0.022* (0.014)
Treated $\times$ FC $\times$ Year(4+)	0.006 (0.004)	0.006 (0.005)	0.013 (0.011)	0.013 (0.015)
Other $\times$ FC $\times$ Year(−1)	0.001 (0.002)	0.002 (0.003)	0.001 (0.006)	0.001 (0.008)
Other $\times$ FC $\times$ Year(0)	−0.001 (0.002)	−0.002 (0.004)	−0.002 (0.007)	−0.002 (0.009)
Other $\times$ FC $\times$ Year(1)	−0.005** (0.002)	−0.006** (0.003)	−0.017*** (0.006)	−0.020** (0.010)
Other $\times$ FC $\times$ Year(2)	−0.004* (0.002)	−0.005 (0.003)	−0.011* (0.006)	−0.014 (0.009)
Other $\times$ FC $\times$ Year(3)	−0.002 (0.002)	−0.002 (0.003)	−0.006 (0.006)	−0.010 (0.010)
Other $\times$ FC $\times$ Year(4+)	−0.001 (0.002)	−0.002 (0.003)	−0.003 (0.006)	−0.002 (0.009)
Treated $\times$ Non-FC	0.012*** (0.002)	0.012*** (0.001)	0.027*** (0.005)	0.026*** (0.004)
Other $\times$ Non-FC	0.001 (0.001)	0.000 (0.001)	−0.000 (0.004)	−0.000 (0.003)
Control Variables	Yes	Yes	Yes	Yes
Plant Fixed Effects	Yes	Yes	Yes	Yes
MSA $\times$ Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	291,358	291,358	291,358	291,358
R^2	0.32	0.32	0.92	0.92

6.4 Tables V–VII: financing constraints and robustness

Read:

- Table V replaces financing constraints with firm-level productivity. The effects do not split by high versus low productivity in the same way, suggesting the baseline split is not merely productivity heterogeneity.
- Table VI uses alternative measures of financing constraints: SA index, credit ratings, debt-to-cash-flow, and investment in excess of cash flow.
- Across alternative measures, treated plants expand and other plants of constrained firms typically contract.
- Table VII uses public versus private status. Other plants of private firms experience negative spillovers, while other plants of public firms do not.

Economic message: The baseline result is not driven by one particular financing-constraint index. It is robust to several alternative proxies and to the public/private distinction.

Table V
Financing Constraints versus Productivity

This table presents variants of the regressions in Table II where *FC* and *Non-FC* are replaced by dummy variables indicating whether firm-level productivity lies below (*Low*) or above (*High*) the median across all firms in the year prior to the treatment. Firm-level productivity is measured using either TFP (columns (1) and (3)) or ROC (columns (2) and (4)). Firm-level TFP is the capital-weighted average of the individual plant-level TFPs across all of the firm's plants, where plant-level TFP is described in Table III. Firm-level ROC is the ratio of total profits—that is, the sum of shipments minus labor and material costs across all of the firm's plants—to total capital stock. All other variables are described in Table II. Standard errors are clustered at the firm level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Investment		Employment	
	TFP (1)	ROC (2)	TFP (3)	ROC (4)
Treated × Low	0.009*** (0.002)	0.010*** (0.002)	0.024*** (0.006)	0.024*** (0.006)
Treated × High	0.010*** (0.002)	0.010*** (0.002)	0.025*** (0.005)	0.026*** (0.005)
Other × Low	−0.001 (0.001)	−0.001 (0.001)	−0.002 (0.005)	−0.002 (0.004)
Other × High	−0.001 (0.001)	−0.001 (0.001)	−0.003 (0.004)	−0.003 (0.004)
Control Variables	Yes	Yes	Yes	Yes
Plant Fixed Effects	Yes	Yes	Yes	Yes
MSA × Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	291,358	291,358	291,358	291,358
R ²	0.32	0.32	0.92	0.92

Table V: financing constraints versus productivity

Table VII
Public versus Private Firms

This table presents variants of the regressions in Table II where the sample includes both public and private firms and *FC* and *Non-FC* are replaced by dummy variables indicating whether the plant belongs to a public or private firm in the year prior to the treatment. Public firms are those covered in Compustat. All other variables are described in Table II. Standard errors are clustered at the firm level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Investment		Employment	
	(1)	(2)	(3)	(4)
Treated	0.009*** (0.002)		0.021*** (0.005)	
Other	−0.001 (0.001)		−0.003 (0.002)	
Treated × Private		0.007*** (0.003)		0.017** (0.007)
Treated × Public		0.010*** (0.002)		0.024*** (0.006)
Other × Private		−0.002* (0.001)		−0.005 (0.003)
Other × Public		−0.000 (0.001)		−0.001 (0.003)
Control Variables	Yes	Yes	Yes	Yes
Plant Fixed Effects	Yes	Yes	Yes	Yes
MSA × Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	838,382	838,382	838,382	838,382
R ²	0.45	0.45	0.95	0.95

Table VII: public versus private firms

Table VI
Alternative Measures of Financing Constraints

This table presents variants of the regressions in Table II where the KZ index and WW index are replaced with alternative measures of financing constraints. In columns (1) and (5), *FC (Non-FC)* is a dummy variable indicating whether the plant belongs to a firm whose SA index (Hadlock and Pierce (2010)) lies above (below) the median across all firms in the year prior to the treatment. In columns (2) and (6), *FC (Non-FC)* is a dummy variable indicating whether the plant belongs to a firm that does not (does) have a credit rating (Compustat item SPDRC) while having long-term debt (item #9) outstanding. In columns (3) and (7), *FC (Non-FC)* is a dummy variable indicating whether the plant belongs to a firm whose debt-to-cash flow ratio—that is, the ratio of long-term debt plus debt in current liabilities (item #34) to income before extraordinary items (item #18)—lies above (below) the median across all firms in the year prior to the treatment. In columns (4) and (8), *FC (Non-FC)* is a dummy variable indicating whether the plant belongs to a firm whose investment in excess of cash flow—that is, capital expenditures (item #30) minus income before extraordinary items divided by total assets (item #6)—lies below (above) the median across all firms in the year prior to the treatment. All other variables are described in Table II. Standard errors are clustered at the firm level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Investment				Employment			
	SA Index (1)	Credit Rating (2)	Debt to Cash Flow (3)	Investment- Cash Flow (4)	SA Index (5)	Credit Rating (6)	Debt to Cash Flow (7)	Investment- Cash Flow (8)
Treated × FC	0.008*** (0.003)	0.008*** (0.002)	0.007*** (0.002)	0.007*** (0.002)	0.019** (0.006)	0.020*** (0.006)	0.019*** (0.006)	0.019*** (0.005)
Treated × Non-FC	0.011*** (0.001)	0.011*** (0.002)	0.012*** (0.002)	0.011*** (0.002)	0.026*** (0.004)	0.028*** (0.005)	0.029*** (0.005)	0.028*** (0.005)
Other × FC	−0.003* (0.002)	−0.002* (0.001)	−0.002* (0.001)	−0.002 (0.001)	−0.007 (0.006)	−0.007* (0.004)	−0.006 (0.004)	−0.006* (0.004)
Other × Non-FC	−0.000 (0.001)	0.000 (0.001)	−0.000 (0.001)	0.000 (0.001)	−0.000 (0.003)	0.001 (0.004)	0.002 (0.004)	0.001 (0.003)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Plant Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MSA × Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	291,358	291,358	291,358	291,358	291,358	291,358	291,358	291,358
R ²	0.32	0.32	0.32	0.32	0.92	0.92	0.92	0.92

6.5 Tables VIII and IX: which other plants are affected?

Read:

- Table VIII shows that spillovers are stronger when the firm has relatively few other plants. If resources needed by the treated plant are spread across fewer plants, each other plant bears a larger reduction.
- Table IX shows that resource withdrawals are concentrated among other plants that are less productive.
- The table also shows stronger withdrawals from plants in peripheral industries rather than main industries.
- Plants farther from headquarters are more likely to lose resources than plants closer to headquarters.

Economic message: Headquarters does not cut resources randomly. It appears to target plants where the opportunity cost of withdrawal is lower, consistent with efficient internal resource allocation.

Table VIII
Few versus Many “Other” Plants

This table presents variants of the regressions in Table II where $Other \times FC$ and $Other \times Non-FC$ are interacted with dummy variables indicating whether the firm's number of “other” plants lies below or above the median across all treated firms in the year prior to the treatment. All other variables are described in Table II. Standard errors are clustered at the firm level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Investment		Employment	
	KZ Index (1)	WW Index (2)	KZ Index (3)	WW Index (4)
Treated $\times$ FC	0.008*** (0.002)	0.008*** (0.003)	0.019*** (0.006)	0.019** (0.008)
Treated $\times$ Non-FC	0.012*** (0.002)	0.011*** (0.001)	0.028*** (0.005)	0.026*** (0.004)
Other $\times$ FC $\times$ (# Other Plants < Median)	-0.004** (0.002)	-0.005** (0.003)	-0.011** (0.006)	-0.014** (0.007)
Other $\times$ FC $\times$ (# Other Plants $\geq$ Median)	-0.001 (0.002)	-0.002 (0.004)	-0.003 (0.006)	-0.002 (0.010)
Other $\times$ Non-FC $\times$ (# Other Plants < Median)	0.000 (0.002)	-0.001 (0.002)	0.001 (0.006)	0.001 (0.005)
Other $\times$ Non-FC $\times$ (# Other Plants $\geq$ Median)	0.001 (0.002)	0.000 (0.001)	-0.000 (0.004)	-0.001 (0.004)
Control Variables	Yes	Yes	Yes	Yes
Plant Fixed Effects	Yes	Yes	Yes	Yes
MSA $\times$ Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	291,358	291,358	291,358	291,358
R ²	0.32	0.32	0.92	0.92

Table IX
Which Other Plants Are Primarily Affected?

This table presents variants of the regressions in Table II where $Other \times FC$ and $Other \times Non-FC$ are interacted with plant characteristics. In Panel A, $Other \times FC$ and $Other \times Non-FC$ are interacted with dummy variables indicating whether the plant's TFP lies below (*Low*) or above (*High*) the median across all of the firm's “other” plants in the year prior to the treatment. In Panel B, $Other \times FC$ and $Other \times Non-FC$ are interacted with dummy variables indicating whether the plant operates in a peripheral or main industry of the firm. Peripheral industries are three-digit SIC industries that account for less than 25% of the firm's total value of shipments in the year prior to the treatment. In Panel C, $Other \times FC$ and $Other \times Non-FC$ are interacted with dummy variables indicating whether the geographical distance between the plant and its headquarters lies below (*Low*) or above (*High*) the median across all of the firm's “other” plants in the year prior to the treatment. Geographical distance is the great-circle distance between the plant's ZIP code and the ZIP code of headquarters. All other variables are described in Table II. The coefficients on $Treated \times FC$ and $Treated \times Non-FC$ are not displayed for brevity. Standard errors are clustered at the firm level. The sample period is from 1977 to 2005. Standard errors are in parentheses (except for the *F*-statistics, where *p*-values are in parentheses). *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Panel A: Plant Productivity			
	Investment		Employment	
	KZ Index (1)	WW Index (2)	KZ Index (3)	WW Index (4)
Other $\times$ FC $\times$ Low	-0.005** (0.002)	-0.006** (0.003)	-0.011** (0.004)	-0.015** (0.006)
Other $\times$ FC $\times$ High	0.001 (0.002)	0.001 (0.003)	-0.002 (0.006)	0.004 (0.009)
Other $\times$ Non-FC $\times$ Low	0.001 (0.002)	-0.001 (0.001)	0.000 (0.005)	-0.001 (0.004)
Other $\times$ Non-FC $\times$ High	0.001 (0.002)	0.000 (0.002)	0.003 (0.005)	-0.000 (0.004)
Other $\times$ FC $\times$ Low versus Other $\times$ FC $\times$ High <i>F</i> -statistic	3.92** (0.048)	3.09* (0.083)	3.31* (0.069)	3.07* (0.080)

(Continued)

Table IX, Panel A: plant productivity

Table IX—Continued

	Panel B: Peripheral versus Main Industries			
	Investment		Employment	
	KZ Index (1)	WW Index (2)	KZ Index (3)	WW Index (4)
Other $\times$ FC $\times$ Main	0.000 (0.002)	-0.001 (0.003)	0.002 (0.006)	0.003 (0.008)
Other $\times$ FC $\times$ Peripheral	-0.005** (0.002)	-0.006** (0.003)	-0.011** (0.006)	-0.017** (0.008)
Other $\times$ Non-FC $\times$ Main	0.001 (0.002)	0.001 (0.002)	0.003 (0.006)	0.002 (0.005)
Other $\times$ Non-FC $\times$ Peripheral	-0.000 (0.002)	-0.001 (0.001)	0.000 (0.005)	-0.002 (0.004)
Other $\times$ FC $\times$ Main versus Other $\times$ FC $\times$ Peripheral <i>F</i> -statistic	3.48* (0.062)	3.03* (0.082)	3.33* (0.068)	3.30* (0.073)
	Panel C: Proximity to Headquarters			
	Investment		Employment	
	KZ Index (1)	WW Index (2)	KZ Index (3)	WW Index (4)
Other $\times$ FC $\times$ Low	0.000 (0.002)	0.001 (0.003)	0.002 (0.006)	0.004 (0.009)
Other $\times$ FC $\times$ High	-0.004** (0.002)	-0.006** (0.003)	-0.012** (0.006)	-0.015* (0.008)
Other $\times$ Non-FC $\times$ Low	0.000 (0.002)	0.000 (0.002)	0.001 (0.005)	0.001 (0.004)
Other $\times$ Non-FC $\times$ High	0.000 (0.002)	-0.001 (0.001)	0.002 (0.005)	-0.001 (0.004)
Other $\times$ FC $\times$ Low versus Other $\times$ FC $\times$ High <i>F</i> -statistic	2.60* (0.094)	3.59* (0.058)	3.08* (0.079)	3.27* (0.071)

Table IX, Panels B–C: peripheral industries and distance to HQ

6.6 Tables X and XI: firm-level investment, employment, and productivity

Read:

- Table X shows that treatment raises firm-level investment and employment.
- Treatment effects are larger for financially unconstrained firms, because they can fund treated-plant expansion without cutting other plants.
- Table XI shows that firm-level TFP and ROC increase after treatment.
- The productivity result supports efficient reallocation rather than mere redistribution within the firm.

Economic message: Firm-level productivity gains are the strongest evidence that internal reallocation creates value rather than simply shifting resources across plants.

Table X
Firm-Level Investment and Employment

The dependent variable is either firm-level investment (columns (1) to (3)) or firm-level employment (columns (4) to (6)). Investment is the ratio of total capital expenditures to total capital stock, where total capital expenditures (total capital stock) is the sum of capital expenditures (capital stock) across all of the firm's plants. Employment is the natural logarithm of the total number of employees across all of the firm's plants. *Treatment* is a dummy variable indicating whether the firm has been treated, that is, a new airline route has been introduced that reduces the travel time between headquarters and one of the firm's plants. *FC (Non-FC)* is a dummy variable indicating whether the firm's measure of financing constraints lies above (below) the median across all firms in the year prior to the treatment. In columns (2) and (5), financing constraints are measured using the KZ index of Kaplan and Zingales (1997). In columns (3) and (6), financing constraints are measured using the WW index of Whited and Wu (2006). The control variables are firm size and firm age, where firm size is the natural logarithm of the sum of shipments across all of the firm's plants, and firm age is the natural logarithm of one plus the number of years the firm has been in the LBD. Both variables are lagged. Standard errors are clustered at the firm level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Investment			Employment		
	(1)	KZ Index (2)	WW Index (3)	(4)	KZ Index (5)	WW Index (6)
Treatment	0.002*** (0.001)			0.004** (0.002)		
Treatment × FC		0.000 (0.001)	0.000 (0.001)		−0.000 (0.002)	0.001 (0.003)
Treatment × Non-FC		0.004*** (0.001)	0.003*** (0.001)		0.009*** (0.002)	0.006*** (0.002)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	33,695	33,695	33,695	33,695	33,695	33,695
R ²	0.41	0.41	0.41	0.88	0.88	0.88

Table X: firm-level investment and employment

6.7 Table XII: other sources of funding

Read:

- Table XII examines whether firms fund expansion using cash, equity, short-term debt, or long-term debt.
- For financially constrained firms, the coefficients are close to zero, consistent with limited access to outside or balance-sheet funding.
- For financially unconstrained firms, there is some evidence of drawing on cash or short-term debt.
- The table reinforces the idea that constrained firms rely on internal real resource reallocation rather than financial slack.

Economic message: Financially constrained firms do not appear to fund treated-plant expansion mainly by issuing securities or drawing down financial assets. They reallocate real resources within the firm.

Table XII
Other Sources of Funding

This table presents variants of the regressions in Table X where the dependent variable is either cash (columns (1) to (3)), equity (columns (4) to (6)), short-term debt (columns (7) to (9)), or long-term debt (columns (10) to (12)). Cash is cash and short-term investments (Compustat item #1) divided by total assets (item #6). Equity is book value of common equity (item #60) divided by total assets. Short-term debt is debt in current liabilities (item #4) divided by total assets. Long-term debt is long-term debt (item #9) divided by total assets. All other variables are described in Table X. Standard errors are clustered at the firm level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	Cash			Equity			Short-Term Debt			Long-Term Debt		
	(1)	KZ Index (2)	WW Index (3)	(4)	KZ Index (5)	WW Index (6)	(7)	KZ Index (8)	WW Index (9)	(10)	KZ Index (11)	WW Index (12)
Treatment	−0.001 (0.001)			−0.000 (0.003)			0.001 (0.001)			0.000 (0.002)		
Treatment × FC		0.000 (0.001)	−0.000 (0.002)		0.000 (0.004)	−0.000 (0.004)		0.000 (0.002)	−0.000 (0.002)		−0.000 (0.003)	−0.000 (0.003)
Treatment × Non-FC		−0.002* (0.001)	−0.002 (0.001)		−0.001 (0.003)	−0.000 (0.003)		0.002 (0.001)	0.002* (0.001)		0.001 (0.003)	0.000 (0.002)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	33,695	33,695	33,695	33,695	33,695	33,695	33,695	33,695	33,695	33,695	33,695	33,695
R ²	0.30	0.30	0.30	0.28	0.28	0.28	0.10	0.10	0.10	0.14	0.14	0.14

7 Short summary

- The paper tests the efficient internal capital markets hypothesis using plant-level Census data.

Table XI
Firm-Level Productivity

This table presents variants of the regressions in Table X where the dependent variable is either firm-level TFP (columns (1) to (3)) or firm-level ROC (columns (4) to (6)). Firm-level TFP and firm-level ROC are described in Table V. All other variables are described in Table X. Standard errors are clustered at the firm level. The sample period is from 1977 to 2005. Standard errors are in parentheses. *, **, and *** denotes significance at the 10%, 5%, and 1% level, respectively.

	TFP			ROC		
	(1)	KZ Index (2)	WW Index (3)	(4)	KZ Index (5)	WW Index (6)
Treatment	0.003*** (0.001)			0.004*** (0.001)		
Treatment × FC		0.002** (0.001)	0.002** (0.001)		0.003** (0.001)	0.003** (0.001)
Treatment × Non-FC		0.005*** (0.001)	0.004*** (0.001)		0.006*** (0.001)	0.005*** (0.001)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	33,695	33,695	33,695	33,695	33,695	33,695
R ²	0.51	0.51	0.51	0.61	0.61	0.61

beneficial. Moreover, Table III shows that this productivity increase comes en-

Table XI: firm-level productivity

- The treatment is a new airline route that reduces travel time between headquarters and one plant.
- Treated plants increase investment, employment, TFP, and ROC.
- Other plants in the same financially constrained firm reduce investment and employment.
- Other plants in financially unconstrained firms do not contract.
- The spillover effect is robust to KZ, WW, SA, credit rating, debt-to-cash-flow, investment-cash-flow, and public/private measures of financing constraints.
- Headquarters withdraws resources more from low-productivity plants, peripheral plants, and plants far from headquarters.
- Firm-level investment, employment, and productivity rise after treatment.
- The evidence supports a value-enhancing internal reallocation view of headquarters.

8 Conclusion

8.1 Core conclusion

Giroud and Mueller show that headquarters reallocates capital and labor within firms when one plant receives a positive investment opportunity shock. This reallocation occurs only when firms are financially constrained, consistent with internal capital markets being most important when external finance is scarce.

8.2 Economic intuition

A reduction in travel time makes the treated plant easier to monitor and more attractive to expand. If the firm is unconstrained, it can fund this expansion without cutting other plants. If it is constrained, headquarters must free resources internally. It does so selectively, reducing investment and employment at other plants where the opportunity cost is lower.

8.3 Takeaway

The paper provides direct plant-level evidence that internal capital markets can be efficient. Headquarters is not merely redistributing resources arbitrarily. It reallocates scarce capital and labor from lower-value plants to higher-value plants, raising firm-wide productivity.